Mengpang Xing

Mengpang's Portfolio (mythventor.github.io)
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Education

Northwestern University — Evanston, IL
Bachelor of Arts in Computer Science and Data Science — *Sep 2023 – Anticipated June 2027*
Relevant Courses: Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Web Development, Database Systems, Distributed Systems

Experience

Cars Commerce (Cars.com) — Chicago, IL
Software Engineer Intern — *Mar 2025 – Aug 2025*

- Architected a production-grade customer data platform from scratch using **React**, **FastAPI**, and **PostgreSQL** to centralize dealership analytics for **1,000+** automotive dealers nationwide.
- Containerized and deployed the application to **AWS ECS** with an automated **Jenkins** pipeline, achieving **200 QPS** with **50ms** average latency and reducing release time by **70%**.
- Migrated infrastructure from **AWS EKS** to **ECS**, eliminating Kubernetes maintenance overhead while preserving high availability for production workloads serving millions in revenue-critical transactions.
- Deployed **16 Metaplane** data monitors across **50+** critical tables, cutting mean time to detect data-quality issues by **40%** and protecting pricing and lead-generation pipelines.
- Modernized internal web platform build system to **PNPM + SWC**, fixing **20** security vulnerabilities and improving build runtime for faster releases.

Y STEM and Chess Inc — Boise, ID
Full-stack Developer Internship — *Jun 2024 – Sept 2024*

- Developed **RESTful APIs** using **Node.js** and **Express.js** to manage user authentication, lesson progression tracking, and chess game state for **500+** students, addressing platform scalability challenges as enrollment grew **3x** year-over-year.
- Designed and implemented **PostgreSQL** database schema to persist student progress, lesson content, and chess game history, optimizing queries to reduce response times by **30%** and enabling teachers to identify struggling students in real-time.
- Built server-side logic for real-time chess move validation and game state synchronization across **100+** concurrent sessions, eliminating data inconsistencies that previously caused **15%** of student sessions to fail.
- Collaborated with design team to deliver responsive, accessible frontend interfaces using **React** and **TypeScript**, reducing student drop-off rates by **25%** through improved lesson interactivity and engagement.
- Tested and debugged full-stack features across **5+** browsers and devices, resolving **15+** critical compatibility issues that were blocking remote learners on older hardware from accessing content.

University of Illinois Chicago — Chicago, IL
STEM Scholar Internship & After-school Program — *Nov 2022 – Aug 2023*

- Collaborated with students from **over 20+ high schools** in a STEM-Focused program, addressing challenges in engineering, architecture, and sustainable development.
- Engineered a Rube Goldberg machine with **12+ mechanical components**, demonstrating proficiency in physics and mechanical engineering principles.
- Designed a community solar microgrid model capable of powering **5 model homes**, exploring sustainable energy practices and electrical engineering applications.

Projects

RubricXpert | *React, Node.js, Flask, OpenAI API (GPT-4o/4o-mini), Python, Longformer, MiniLM* — Mar 2025

- Built backend of a parallelized AI essay evaluation system using **Longformer**, **MiniLM**, and **GPT-4o/4o-mini** with rubric-aligned grading, file conversion, theme extraction, and meta-analysis via **ThreadPoolExecutor** with **10 concurrent workers**.
- Implemented **GPU-optimized Longformer** execution with robust error handling for scalable, real-time feedback that outperformed services like Cograder; processed **50+ criteria** per essay with criterion-level scores.
- Deployed full-stack web application using **Node.js**, **React**, **Flask**, and **HTML/CSS**, producing consistent machine-readable **JSON** outputs for downstream analysis and dashboards (rubricxpert.org pending).

Weather Application | *React, Tailwind CSS, Open-Meteo API, Geolocation API, JavaScript* — Aug 2025

- Built comprehensive weather platform with **20+ meteorological parameters** including solar radiation and soil temperature using **React** and **RESTful API** integration—providing professional-grade insights beyond typical weather apps.
- Implemented **GPS detection**, global geocoding with **Geolocation API**, and **metric/imperial unit switching** with automatic location formatting and responsive, accessible interface featuring dark/light themes using **Tailwind CSS**.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML, CSS, DrRacket
Frameworks & Libraries: React, Node.js, Flask, FastAPI, Express.js, Vue.js, Django, TailwindCSS, Spring Boot
Cloud & DevOps: AWS (ECS, EKS, EC2, S3, SES), Docker, Jenkins, CI/CD, Git, GitHub
Databases & Tools: PostgreSQL, MySQL, Redis, RESTful APIs, OpenAI API, JWT, OAuth
AI/ML: GPT-4o, Longformer, MiniLM, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy